SOFTWARE ARCHITECTURE PROJECT

Job Recruitment and Internship Platform

A platform that helps students and job seekers find jobs and internship opportunities.

TEAM NAME	COURSE	MILESTONE
Best Project Makers	Software Architecture	01

TEAM MEMBERS

No.	Team Member	Student ID
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GitHub Repository

https://github.com/Mostafamohamed-11/Job-Recruitment-and-Internship-Platform.git

Milestone 1 submission • Prepared by Best Project Makers

TEAM CONTRIBUTIONS

The following table summarizes each team member's assigned responsibilities for Milestone 1.

Team Member	Assigned Tasks
Ahmed Khorshed	Task 1 — System Overview
Bassel Osama	Task 2 — Stakeholders and Concerns
Mostafa Mohamed	Task 3 — System Context Task 4 — Functional Requirements
Yassin Sherif	Task 5 — Quality Attribute Scenarios
Marwan Abdel moniem	Task 6 — Constraints
Marsellino Emad	Task 7 — Architecturally Significant Requirements (ASRs)

PROJECT SUMMARY

The Job Recruitment and Internship Platform is designed to help students and job seekers discover job openings and internship opportunities in one place.

System Overview

Job Recruitment and Internship Platform

1.1 Purpose

The Job Recruitment and Internship Platform is a web platform that connects job seekers and employers. It supports four main functions: job applications, CV management, candidate matching, and interview scheduling.

1.2 The Problem

Hiring today is slow and scattered:

- Applicants must search many websites and send their CVs again and again.
- Recruiters receive many CVs in different formats and must read them one by one.
- Applicants often get no confirmation or news about their application.
- CVs contain personal data, so they must be kept safe.

This platform puts the whole process in one place.

1.3 Who Uses It

User	What they do
Applicant	Searches for jobs and internships, uploads a CV, and applies
Recruiter	Publishes vacancies, reviews candidates, and schedules interviews
Platform Administrator	Checks job listings and manages users to keep the platform safe

1.4 External Systems

The platform depends on two outside services:

System	What it does for the platform
Email Service	Sends application confirmations and interview notifications
Apache Tika	Reads uploaded CV files and extracts the text, so the platform can match candidates to jobs

1.5 Why It Is Architecturally Interesting

The system looks simple, but it must solve several design challenges at the same time. Each one links to a quality attribute scenario in Section 5.

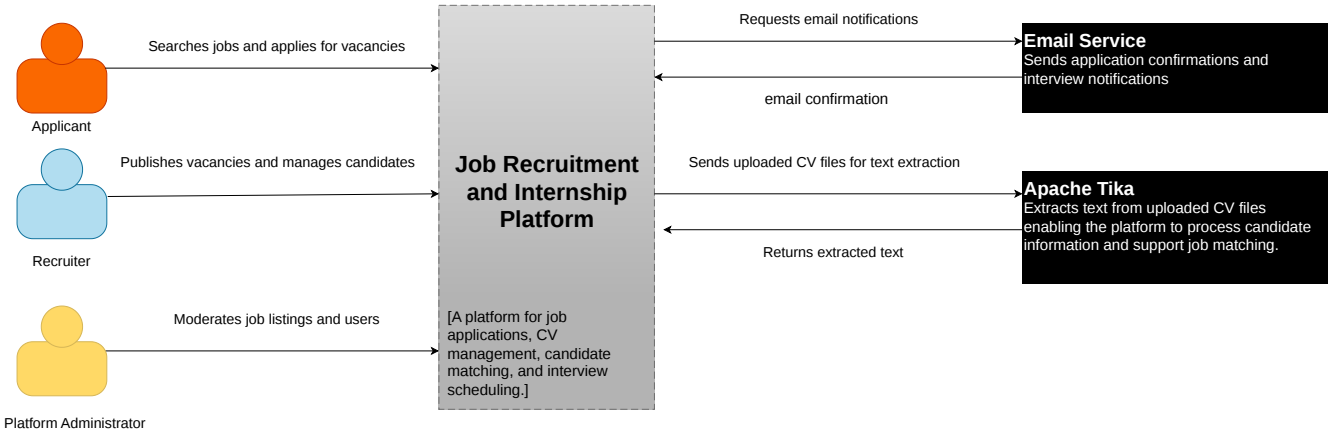
Challenge	Why it matters	Scenario
Fast search	Applicants search through thousands of vacancies and expect quick results	Performance (QAS-1)
Outside service failure	If the Email Service is down, applications must still be saved and emails sent later	Availability (QAS-2)
Data protection	CVs and candidate lists are private and must be safe from unauthorized access	Security (QAS-3)
Easy change	Apache Tika should be replaceable without changing the rest of the system	Modifiability (QAS-4)
Traffic peaks	Events like a university job fair can raise users from 500 to 5,000	Scalability (QAS-5)
Easy to use	First-time applicants must be able to apply quickly without help	Usability (QAS-6)

These challenges pull the design in different directions. For example, strong security checks can slow down search, and extra servers for peak traffic add cost. The architecture must balance them, and this makes the platform a good subject for architectural study.

2. Stakeholders and Concerns

#	Stakeholder	Main Concerns
1	**Applicant**	The applicant wants to find a good job or internship, upload their CV easily, and know if the company accepted or rejected their application. They also want their personal information to stay safe.
2	**Recruiter**	The recruiter wants to post job vacancies, find people with the right skills, check their CVs, and arrange interviews without wasting too much time.
3	**Platform Administrator**	The admin wants to make sure everything is working well, remove fake job posts, manage users, and solve any problems that happen on the platform.
4	**Email Service**	The email service should send notifications on time, like when someone applies for a job or gets an interview invitation. Emails should not fail or arrive late.
5	**CV Parsing API**	The CV parsing system should read CVs correctly and find important information like skills, education, and experience. It should also work with different CV formats.
6	**Hiring Company**	The company wants to find suitable people for its jobs, save time during the hiring process, and make sure recruiters can manage applications easily.

3. System Context



View the interactive system context diagram

4. Functional Requirements

ID	User Story	prioritized using MoSCoW	Will implemented in Prototype?
US01	As an Applicant, I want to register and log in so that I can access the platform securely.	Must Have	Yes
US02	As an Applicant, I want to browse and search available jobs so that I can find suitable opportunities.	Must Have	Yes
US03	As an Applicant, I want to apply for a job and upload my CV so that recruiters can review my application.	Must Have	Yes
US05	As a Recruiter, I want to create, edit, and delete job vacancies so that I can manage recruitment opportunities.	Must Have	Yes
US06	As a Recruiter, I want to view applicants and their submitted CVs so that I can evaluate candidates.	Must Have	Yes
US07	As a Recruiter, I want to edit or delete my job posts so that I can manage vacancies.	Should Have	No

ID	User Story	prioritized using MoSCoW	Will implemented in Prototype?
US08	As a Platform Administrator, I want to manage user accounts so that I can control access to the platform.	Must Have	No
US09	As an Applicant, I want to receive an email after applying so that I know my application was submitted.	Could Have	No
US10	As an Applicant, I want to filter jobs by type so that I can find internships or full-time jobs.	Could Have	No
US11	As a Recruiter, I want to reject unsuitable applications so that I can manage the recruitment process.	Should Have	No

Job Recruitment and Internship Platform

Section 5: Quality Attribute Scenarios

5.1 Introduction

This section describes six quality attribute scenarios for the Job Recruitment and Internship Platform. Each scenario uses the six-part form: source, stimulus, artifact, environment, response, and response measure. The scenarios are based on the main features of the platform: job applications, CV management, candidate matching, and interview scheduling. They cover the needs of its users (applicants, recruiters, and administrators) and the services it depends on (email notifications and CV text extraction).

5.2 Overview

ID	Quality Attribute	Main Concern	Related Users / Services
QAS-1	Performance	Fast job search	Applicant
QAS-2	Availability	Email failure must not stop applications	Email Service
QAS-3	Security	Protect CVs and candidate data	Applicant, Recruiter
QAS-4	Modifiability	Replace the CV text-extraction tool easily	Apache Tika
QAS-5	Scalability	Handling a sudden rise in users	All users
QAS-6	Usability	Easy first-time application	Applicant

5.3 Scenarios

QAS-1: Performance (Job Search)

Part	Description
Source	Applicant
Stimulus	Searches for jobs using a keyword and filters (location, field)
Artifact	Job search function
Environment	Normal operation, 200 concurrent users, 50,000 active vacancies
Response	The platform returns matching vacancies, sorted by relevance
Response Measure	95% of search requests complete in under 500 ms, and 99% in under 1 second

QAS-2: Availability (Email Service Failure)

Part	Description
Source	Email Service
Stimulus	The service becomes unavailable while applicants submit applications
Artifact	Notification module
Environment	Normal operation, 100 applications submitted per hour
Response	The platform saves every application, places the emails in a queue, and retries sending automatically
Response Measure	0% of applications are lost, and 99% of emails are sent within 15 minutes after recovery.

QAS-3: Security (Unauthorized Access to CVs)

Part	Description
Source	Malicious user (logged-in applicant or unauthenticated user)
Stimulus	Tries to open another applicant's CV or a recruiter's candidate list by changing the URL or ID
Artifact	CV storage and candidate management modules
Environment	Normal operation, over the internet
Response	The platform denies access (HTTP 403), records the attempt in the audit log, and alerts the administrator after 5 failed attempts
Response Measure	100% of 500 test attacks are blocked and logged within 1 second.

QAS-4: Modifiability (Replacing Apache Tika)

Part	Description
Source	Developer
Stimulus	Wants to replace Apache Tika with another CV text-extraction tool
Artifact	CV processing component (Tika integration)
Environment	Design time, during a planned release
Response	The developer changes only the CV adapter. Matching, search, and notification modules stay unchanged
Response Measure	Change is completed in 3 person-days or less. Only 1 component is modified. 100% of regression tests pass after the change

QAS-5: Scalability (Job Fair Traffic Spike)

Part	Description
Source	Applicants and recruiters
Stimulus	Traffic grows from 500 to 5,000 concurrent users during a university job fair
Artifact	Web and application servers
Environment	Peak load, normal operation
Response	The platform adds server instances automatically and balances the load between them
Response Measure	Extra capacity is ready within 5 minutes. 95% of requests complete in under 1 second. Error rate stays below 1%

QAS-6: Usability (First-Time Applicant)

Part	Description
Source	First-time applicant
Stimulus	Wants to apply for an internship and upload a CV
Artifact	Application and CV upload screens
Environment	Normal operation, on a laptop or mobile browser
Response	The platform guides the user step by step and shows a confirmation message after submission
Response Measure	90% of 10 test users finish the application in under 5 minutes without help.

Constraints Analysis

Job Recruitment and Internship Platform

1. Introduction

This section identifies the technical, business and organizational constraints of the Job Recruitment and Internship Platform, including budget, technology, integrations and deadlines. It is based on the Level 1 context diagram and the MoSCoW-prioritized user stories table.

The platform has three actors (Applicant, Recruiter, Platform Administrator) and two external systems (Email Service and Apache Tika). Main features are CV uploads, job matching, application tracking, interview scheduling and notifications.

Values that are not yet known are marked [confirm] and highlighted in yellow. Replace them with the team's real numbers.

2. Technical Constraints

ID	Constraint	Description / Impact	Source
T1	Apache Tika is the only CV text extractor	CV files must be sent to Tika and the extracted text returned. Job matching depends on Tika being available. Poor extraction from scanned or image-only PDFs lowers matching quality.	Context diagram
T2	Email Service is an external dependency	Sends application confirmations and interview notifications. The platform cannot control delivery; emails may be delayed, fail, or hit rate limits on free plans.	Context diagram
T3	CV file restrictions	Accepted formats (PDF, DOCX), a maximum file size (e.g. 5 MB) and a virus/malware check are needed. Storage must be planned for.	US03, US06
T4	Security and authentication	Hashed passwords, role-based access (Applicant, Recruiter, Admin) and HTTPS. CVs contain personal data, so access must be restricted.	US01, US08
T5	Prototype scope limited to 5 stories	Only US01, US02, US03, US05 and US06 are built. Email, filtering, rejection and admin features are not implemented. The prototype must run end to end without them.	User stories table
T6	Matching must stay simple	Keyword-based matching on Tika text is realistic. AI or ML ranking is out of scope for the prototype because of limited time and data.	Project scope
T7	Performance and availability	Search should respond quickly, and CV extraction should be asynchronous so uploads do not block users. Tika can be slow on large files.	Non-functional
T8	Browser and device support	Must work on current desktop and mobile browsers (responsive web app). Applicants often use phones.	Non-functional

3. Business Constraints

ID	Constraint	Impact
B1	Limited budget	Student project, approx. [confirm]. Only free or open-source tools are used: Apache Tika, free-tier hosting and a free-tier email service. Rules out paid CV-parsing APIs and paid hosting.
B2	Fixed deadline	Deadline: [confirm date]. Forces MoSCoW prioritization: Must Haves first, Could Haves postponed.
B3	Data protection and privacy	CVs and personal data fall under privacy laws (e.g. GDPR or local equivalents). Requires user consent, deletable data, a privacy notice and limited retention.
B4	Fair recruitment	Matching and filtering must not discriminate, so the system should show suitability rather than auto-reject. Affects job matching and US11.
B5	Listing quality	Fake or inappropriate vacancies hurt trust, so moderation by the Administrator is needed. US08 is a Must Have but is excluded from the prototype, which is a business risk to document.
B6	Both sides must benefit	The platform needs enough applicants and recruiters to be useful, and no monetization model is defined for the prototype. Out of scope now but affects long-term viability.

4. Organizational Constraints

ID	Constraint	Impact
O1	Small student team	Team size: [confirm number] members, with limited availability alongside other courses. Limits parallel work and the number of features.
O2	Mixed skill levels	Not everyone may know the chosen stack, Tika or email APIs. Needs task allocation by strength and some learning time.
O3	Coordination	Work must be shared through Git, a task board and regular meetings, with a defined role for each member. Reduces merge conflicts and duplicated work.
O4	Academic requirements	Deliverables must follow course guidelines (diagrams, user stories, documentation, presentation). Documentation takes time away from coding.
O5	No real users or stakeholders	Requirements come from the team and assumptions. Feedback comes late, so requirements may change.
O6	Dependency on third-party services	Team members cannot fix Tika or email provider downtime. Needs fallbacks such as mocks or a local Tika.

5. Integration Constraints

Integration	Direction	Constraint
Apache Tika	Platform sends CV, receives extracted text	Must be deployed (local or Docker). Network latency, file-size limits, text-only output.
Email Service	Platform requests notification, receives confirmation	Needs API key or SMTP setup. Free-tier quotas. A "sent" status is not a guarantee of delivery.

6. Budget and Deadline Overview

Area	Planned approach
Budget	Near-zero: open-source stack, free hosting and email tier.
Phase 1	Must Haves in the prototype (US01, US02, US03, US05, US06).
Phase 2	Should Haves (US07, US11).
Phase 3	Could Haves (US09, US10) and Admin features (US08).
Risk if limits are exceeded	Drop Could Haves first, then Should Haves. Never drop Must Haves in the prototype.

7. Constraint Risks and Mitigation

The table links the main constraints to the risks they create and how the team can reduce them.

ID	Risk	Related constraints	Mitigation
R1	Tika unavailable or poor extraction	T1, O6	Run Tika locally in Docker; fall back to manual keyword entry or a mock extractor.
R2	Email provider limits or downtime	T2, O6	Use a mock email service in the prototype; email is Could Have (US09).
R3	Deadline missed	B2, O1, O4	Follow MoSCoW; drop Could Haves, then Should Haves.
R4	Privacy breach of CV data	T4, B3	Role-based access, HTTPS, consent on upload, deletion option.
R5	Unmoderated fake listings	B5, T5	Admin actions done directly in the database for the prototype; document the risk.

8. Assumptions

- Budget is near zero; only free or open-source tools are used.
- The prototype is a responsive web application implementing US01, US02, US03, US05 and US06.
- Job matching is keyword-based and uses the text returned by Apache Tika.
- Email notifications are shown in the architecture but not built in the prototype.
- Requirements come from the team, as there are no real stakeholders.

9. Issues Found in the Project Documents

While reading the context diagram and user stories, the following inconsistencies were noticed. They should be fixed or justified before submission.

#	Issue	Detail	Suggested action
1	US07 duplicates US05	Both cover editing and deleting job posts.	Merge them or reword one.
2	US04 is missing	Numbering jumps from US03 to US05.	Check whether it was removed and fix the numbering.

#	Issue	Detail	Suggested action
3	US08 (admin) is Must Have but not in the prototype	The teacher may ask why.	Justify in the report, e.g. admin actions are done directly in the database for the prototype.
4	Interview scheduling and job matching have no user story	Both appear in the project idea.	Add stories or state they are out of scope.
5	Email integration vs US09	The diagram shows "Requests email notifications" but US09 is Could Have and not implemented.	Explain that the integration exists in the architecture but is not built in the prototype.

10. Conclusion

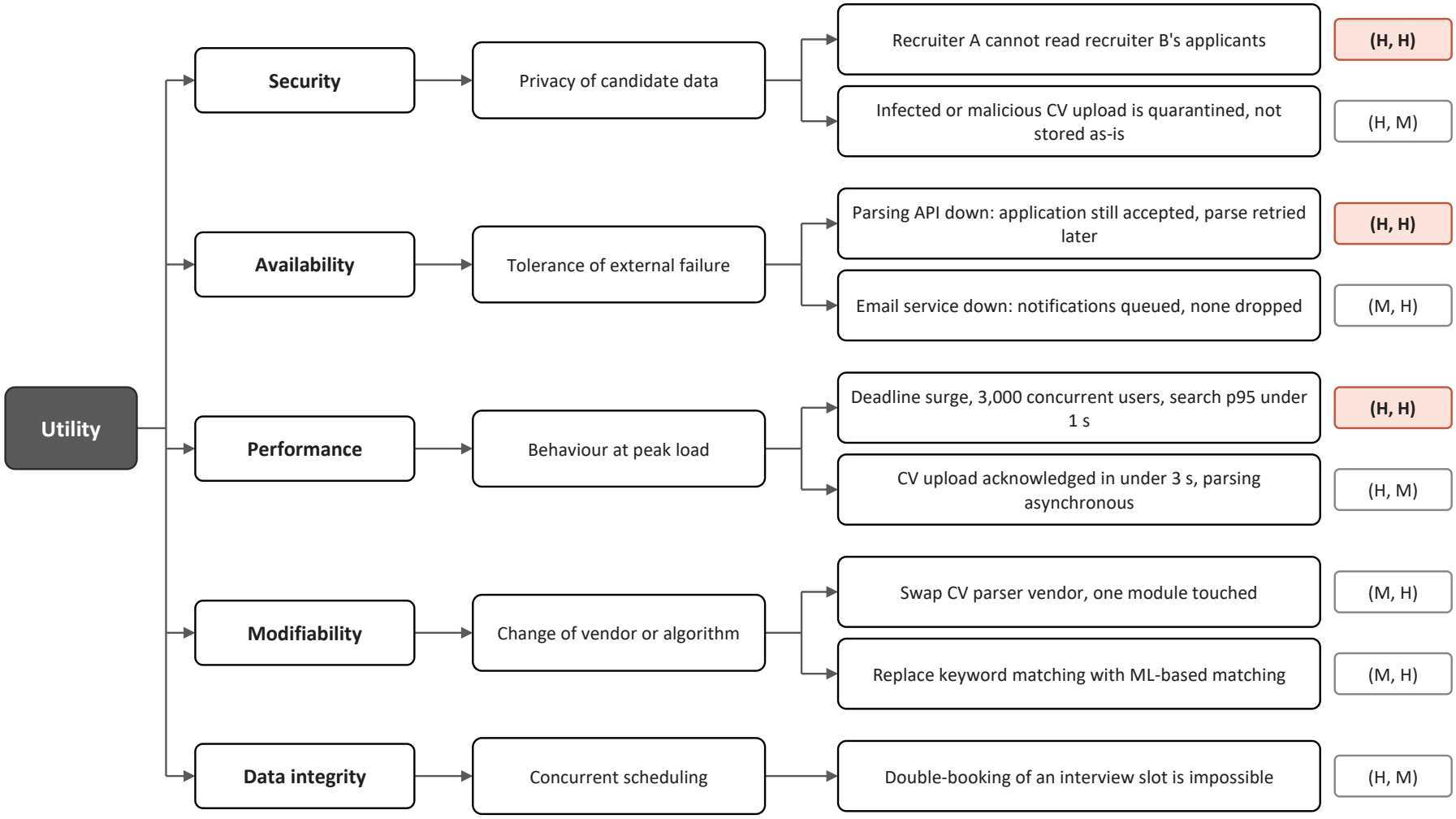
The project is mainly limited by a small student team, a fixed deadline, a near-zero budget and its dependence on two external services (Apache Tika and the Email Service). Prioritizing Must Have stories, keeping matching simple, using open-source tools and planning fallbacks for external services keep the prototype realistic and deliverable.

7. Architecturally Significant Requirements (ASRs)

Job Recruitment and Internship Platform

Candidate ASRs

ID	Requirement	Quality attribute
ASR-1	A recruiter can only see the CVs and applications for their own vacancies. Candidate data is encrypted when it is sent and when it is stored. Every uploaded CV is scanned for viruses, and every time someone opens a candidate record it is written to a log.	Security
ASR-2	When 3,000 people are on the platform at once, job search should answer in under 1 second for 95% of requests. A CV upload should be confirmed in under 3 seconds.	Performance/ Scalability
ASR-3	If the email service goes down, the platform still accepts applications and loses no data. The system keeps retrying parsing and emails for up to 24 hours.	Availability
ASR-4	Changing the CV parsing vendor or the matching algorithm should only affect one module. The application, recruiter, and scheduling code stays untouched.	Modifiability
ASR-5	Two recruiters can never book the same time slot with a candidate, and a candidate can never accept two interviews at the same time, even if the requests arrive together.	Data integrity
ASR-6	When an admin removes a job listing, everyone should stop seeing it within 60 seconds. Every moderation action is recorded with the name of the admin who did it.	Auditability
ASR-7	The main applicant tasks (searching for jobs, applying, and tracking an application) should work well on a phone browser, even with a slow connection.	Usability


Utility
Security
Privacy of candidate data
Recruiter A cannot read recruiter B's applicants
(H, H)
Infected or malicious CV upload is quarantined, not stored as-is
(H, M)
Availability
Tolerance of external failure
Parsing API down: application still accepted, parse retried later
(H, H)
Email service down: notifications queued, none dropped
(M, H)
Performance
Behaviour at peak load
Deadline surge, 3,000 concurrent users, search p95 under 1 s
(H, H)
CV upload acknowledged in under 3 s, parsing asynchronous
(H, M)
Modifiability
Change of vendor or algorithm
Swap CV parser vendor, one module touched
(M, H)
Replace keyword matching with ML-based matching
(M, H)
Data integrity
Concurrent scheduling
Double-booking of an interview slot is impossible
(H, M)